July 2025 CSE 220
Online on Convolution
Subsections: A1, A2
Total Marks: 10 Total Time: 30 Minutes

January 28, 2026

In Offline 1, you computed the output of a discrete-time LTI system using the **impulse response** $h[n]$. In this assignment, you will be given the **step response** $s[n]$ of an LTI system and must compute the output for any input $x[n]$.

Use the identities given below.

$$h[n] = s[n] - s[n-1] \quad (s[-1] = 0),$$
$$\Delta x[n] = x[n] - x[n-1] \quad (x[-1] = 0),$$
$$y[n] = (\Delta x * s)[n].$$

Files provided

`step_response.txt` (contains $s[n]$), `input_signal.txt` (contains $x[n]$).

Format

Line 1: n_{start} n_{end}, Line 2: samples for $n = n_{start} \ldots n_{end}$.

Tasks

1. Read $s[n]$ from `step_response.txt` and construct a `Signal` object.
2. Compute and plot the recovered impulse response $h[n] = s[n] - s[n-1]$.
3. Read $x[n]$ from `input_signal.txt`. Compute $\Delta x[n] = x[n] - x[n-1]$ using only your `Signal` operations.
4. Compute the output using **only** the step response: $y_s[n] = (\Delta x * s)[n]$. **You must reuse your Offline 1 LTI convolution code. Do not use `numpy.convolve`.**
5. Verify using the recovered impulse response: $y_h[n] = (x * h)[n]$. Show that both outputs $y_s[n]$ and $y_h[n]$ match closely.

Marks Distribution

- Reading + plotting $s[n]$: 1
- Determining and plotting $h[n]$: 3
- Computing $\Delta x[n]$: 2
- Output using step response (y_s): 3
- Verification (showing outputs match): 1

Submission:

Submit `{student_id}.py` to moodle. If you have used multiple Python files then you should zip all of them together and upload to moodle.